# Explainable Cyberbullying Detection Using Deep Learning and Transformer Models with LIME

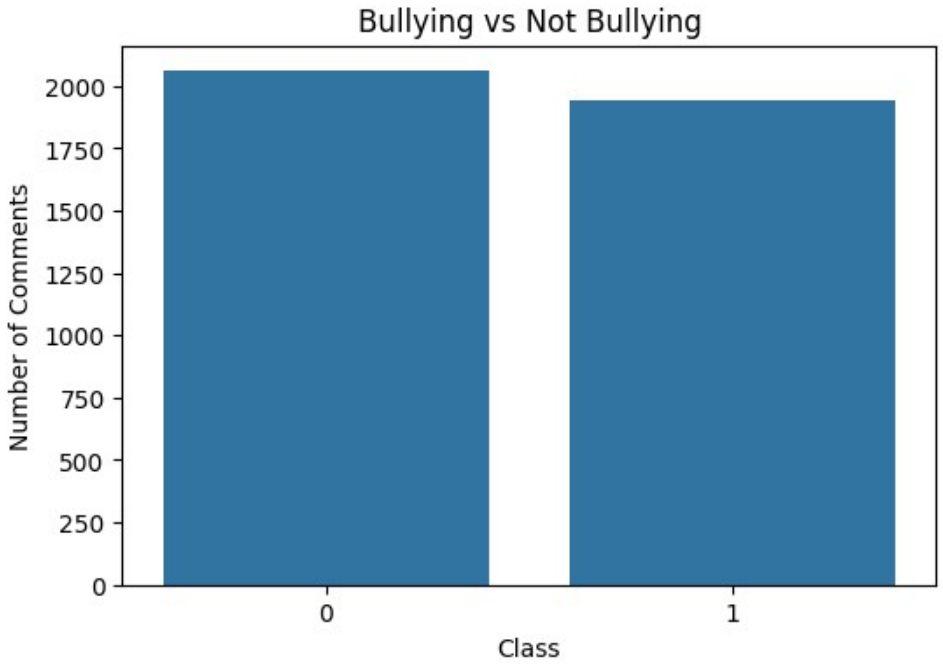
## A Binary Detector (RoBERTa) with Word-Level Explanations


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- Department of Computer Science and Engineering
- Premier University, Chattogram • October 2026

# The Problem & Why It Matters

Speaker 1

- **Millions of comments are posted daily on YouTube, Reddit, Facebook.**
- A large share is hostile — and manual moderation cannot keep up.
- **Most existing detectors give only a label, no reason.**
- A black-box moderation decision is hard to trust or appeal.
- **Our goal: high accuracy + a clear explanation for every decision.**



# Objectives & Contributions

Speaker 1

- **Build a binary detector (Bullying vs Not-Bullying) on multi-platform data.**
- Compare four models: Logistic Regression, LSTM, GRU, RoBERTa.
- **Beat the base paper (Esan et al., 77%) — on our data AND on their own data.**
- **Add LIME so every prediction can be explained word by word.**
- Give an honest multi-dataset comparison against two reference studies.

# Related Work — Where We Fit

Speaker 1

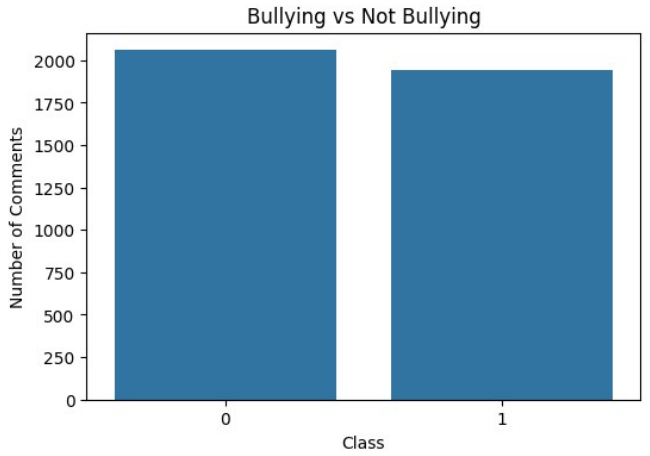
| Study | Best Model | Accuracy | Explainable |
|---|---|---|---|
| Esan et al. (2024) [base] | LSTM | 77.00% | No |
| Ogunleye & Dharmaraj (2023) | RoBERTa | 87.00% | No |
| Muneer et al. (2023) | BERT+biLSTM+CNN | 97.40% | No |
| Alrowais (2024) | RoBERTa + GloVe | 95.00% | No |
| Sihab-Us-Sakib (2024) | XLM-RoBERTa | 82.61% | No |
| Azhar & Runa (2025) | RoBERTa | 83.90% | No |
| Our Work | RoBERTa + LIME | 92.33% | Yes |

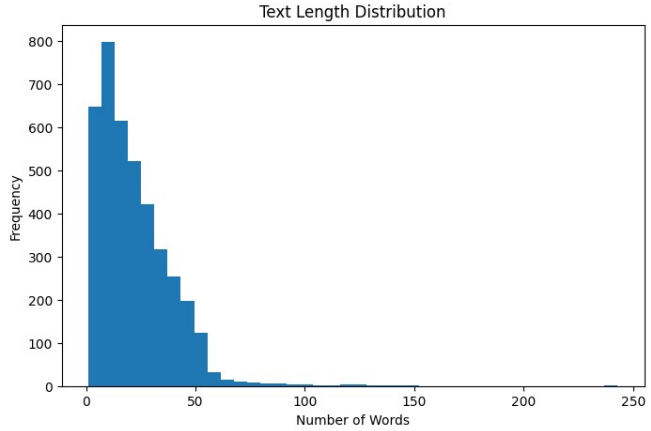
**We reviewed 12 studies — almost none explain their decisions. That gap is our contribution.**

# Dataset

Speaker 1

- **~4,000 English comments**
- 1,000 self-collected: YouTube, Reddit, Facebook
- 3,000 from Jigsaw toxic-comment corpus
- **Near-balanced: 2,059 Not-Bullying / 1,941 Bullying**
- 0 = Not-Bullying, 1 = Bullying
- Avg length ~22.6 words




Left: class balance. Right: comment-length distribution.

# Dataset Examples (6 Bully Types)

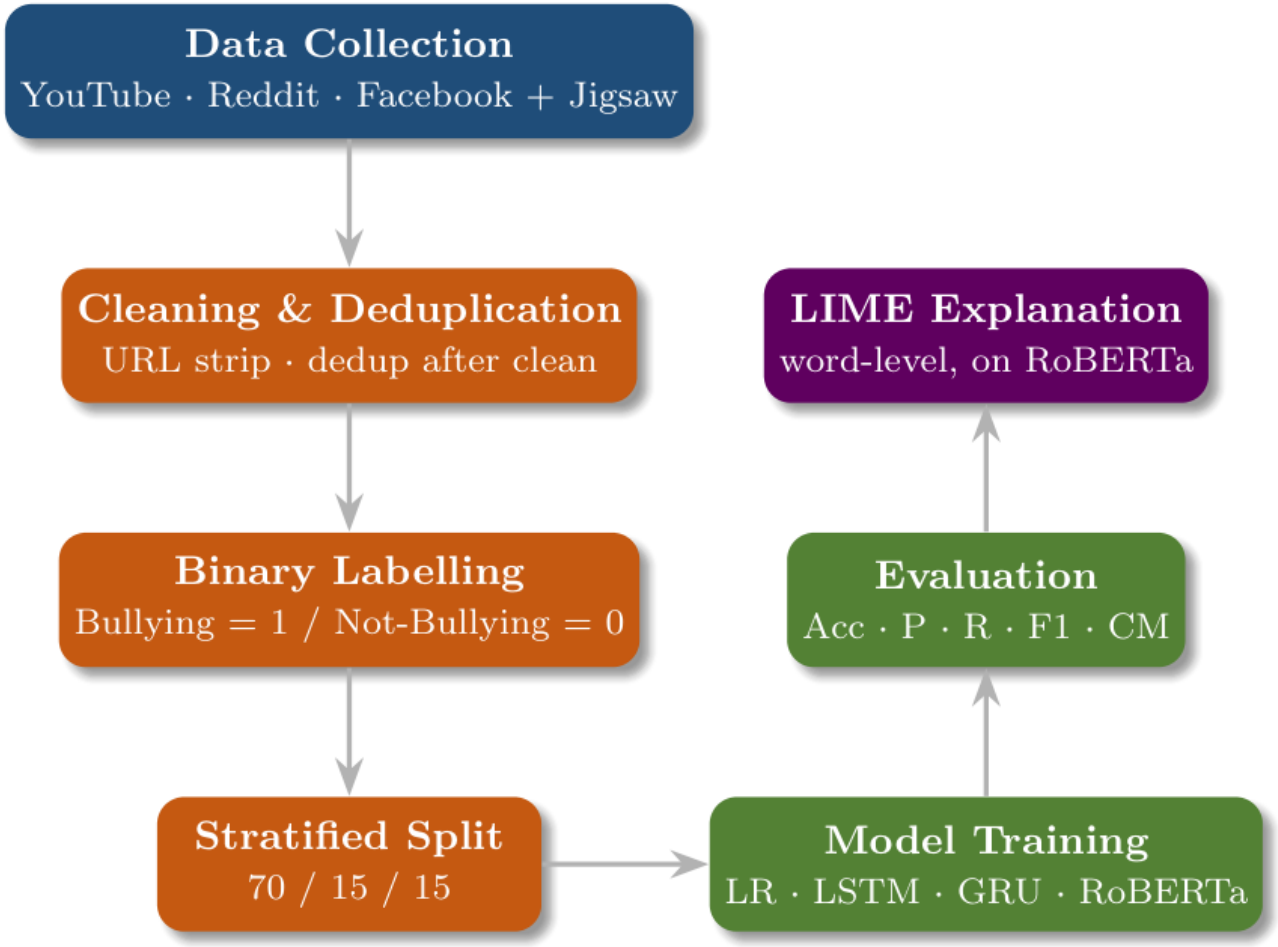
Speaker 1

| Comment | Label | Type |
|---|---|---|
| Show this video and Ronaldo's stats to him. | Not-Bullying | -- |
| Bolton is a clown | Bullying | Troll |
| Religion is primitive and a big waste of time. | Bullying | Religion |
| death to the infidel, whoever that may be | Bullying | Threat |
| This is what happens when you elect a moron. | Bullying | Political |
| Mass immigration is destroying the West. | Bullying | Racism |

**Strong wording alone is not bullying — the task needs context, not keywords.**

# Methodology — The Pipeline

Speaker 1



# Models & Why LIME



- **Four models, simple → advanced:**
  - TF-IDF + Logistic Regression (baseline)
  - LSTM and GRU (read word order)
  - **RoBERTa (pretrained transformer)**
- **LIME = model-agnostic, local, word-level, lightweight**




LIME perturbs the comment, watches RoBERTa's probability, and scores each word.

# Results — RoBERTa Wins

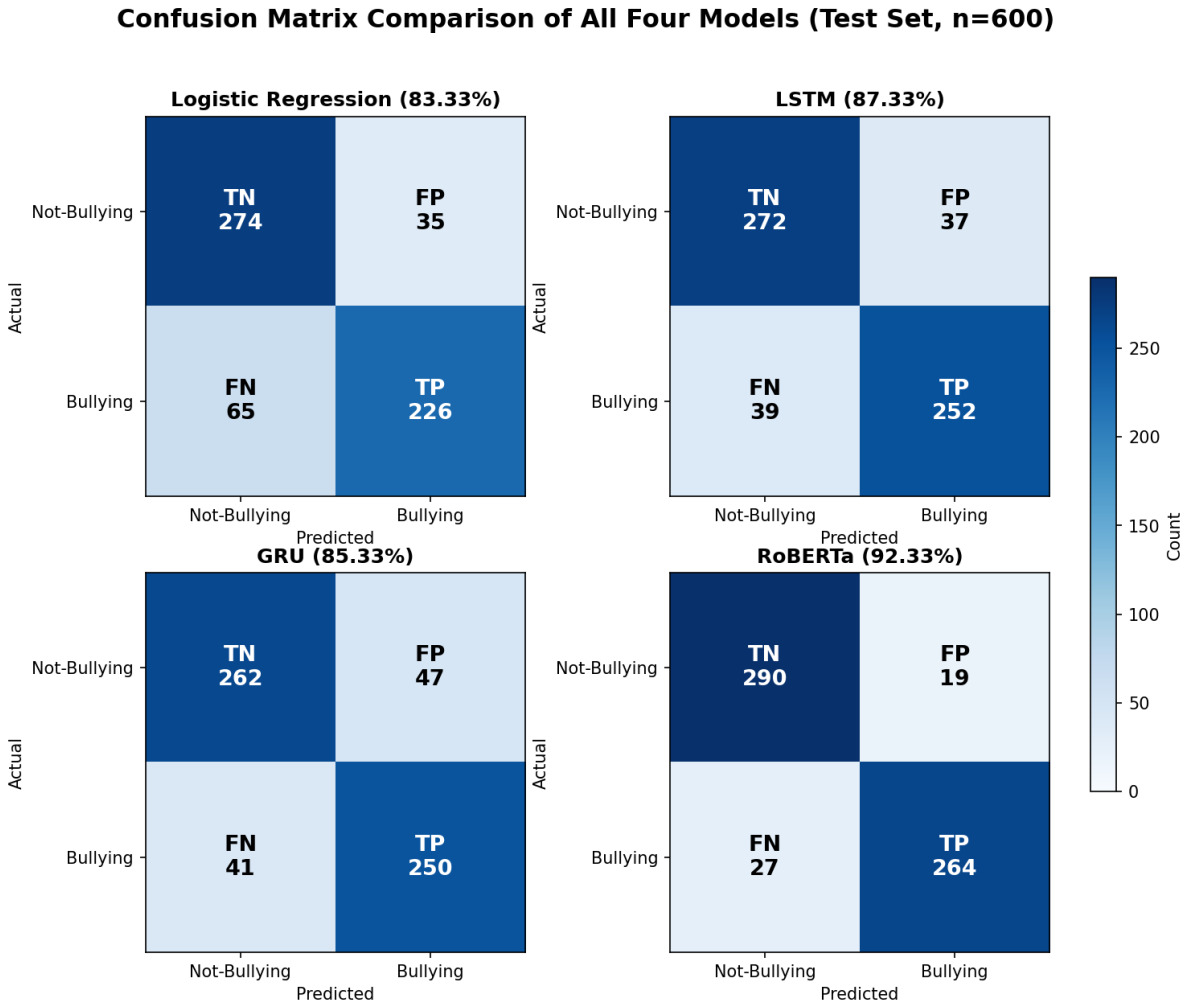
Speaker 2

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Logistic Regression | 83.33% | 83.62% | 83.33% | 83.27% |
| GRU | 85.33% | 85.36% | 85.33% | 85.34% |
| LSTM | 87.33% | 87.33% | 87.33% | 87.33% |
| RoBERTa (our model) | 92.33% | 92.36% | 92.33% | 92.33% |

**RoBERTa: 92.33% — 5 points above the best recurrent model, and far above the base paper's 77%.**

# Confusion Matrices — All Four Models

Speaker 2



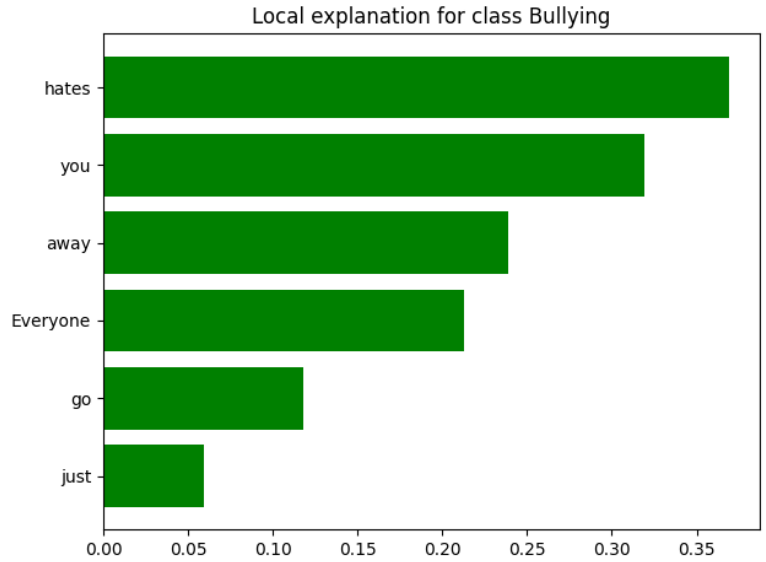
# LIME — Explaining Correct Predictions

Speaker 2

```
Comment:
Everyone hates you, just go away

Binary Prediction: Bullying
Binary Confidence: 93.30%

Important Words:
------------------------------------------------
hates        +0.3688  -> Bullying
you          +0.3189  -> Bullying
away         +0.2388  -> Bullying
Everyone     +0.2127  -> Bullying
go           +0.1177  -> Bullying
just         +0.0595  -> Bullying
------------------------------------------------
```




"Everyone hates you, just go away" → Bullying (93.30%). LIME shows the words that drove it: hates, you, away. The decision is transparent and defensible.

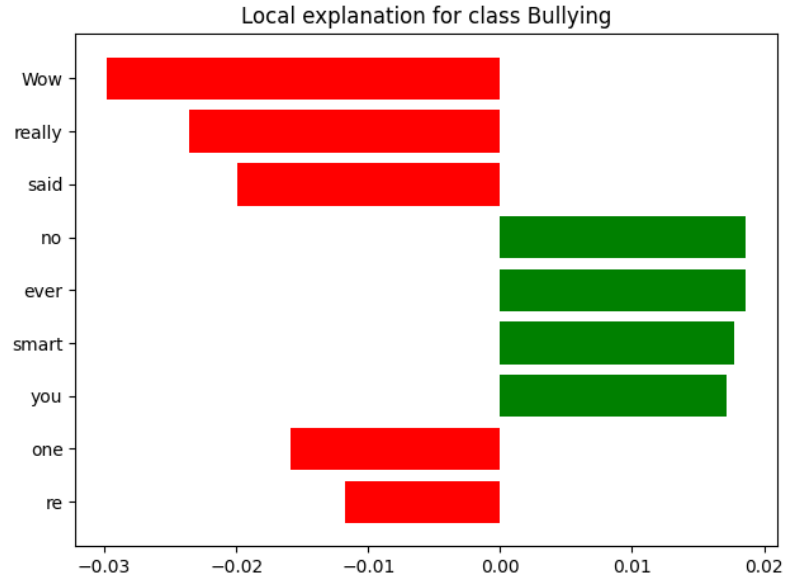
# Honest Limitation — Sarcasm

Speaker 2

```
Comment:
Wow, you're really smart... said no one ever.

Binary Prediction: Not-Bullying
Binary Confidence: 95.32%

Important Words:
------------------------------------------------
Wow          -0.0298  -> Not-Bullying
really       -0.0236  -> Not-Bullying
said         -0.0200  -> Not-Bullying
no           +0.0186  -> Bullying
ever         +0.0186  -> Bullying
smart        +0.0177  -> Bullying
you          +0.0171  -> Bullying
one          -0.0159  -> Not-Bullying
re           -0.0118  -> Not-Bullying
------------------------------------------------
```




"Wow, you're really smart… said no one ever." → predicted Not-Bullying (WRONG). Positive words (Wow, really, said) fool the model. LIME makes this failure visible — a strength of explainability.

# Social Impact & Conclusion

Speaker 2

- Protects vulnerable users; reduces moderator burden.
- **Explainable decisions → fairer, reviewable moderation.**
- Best used human-in-the-loop (assist, not replace).
- **Conclusion: RoBERTa best in every setting + LIME adds trust.**
- Beats the base paper (77%) on our data (92.33%) and theirs (89.01%).

# Future Work & Thank You

Speaker 2

- Close the training gap (reference code hit 95.25% on our data).
- Make the model sarcasm-aware.
- Bring back fine-grained bully types.
- **Multilingual: Bangla + English + Hindi, with type.**

**Thank You — Questions?**